
Project Module 4

Privacy Risk & Privacy-Aware Design

Course CSCE 4930: Ethical AI
Instructor Dr. Alia El Bolock
Dataset German Credit Dataset
Model Decision Tree (max_depth=5, Gini, min_samples_leaf=10)
Task Predict creditworthiness: Good (1) or Bad (0)

Name	ID
Adham Ali	900223243
Omar Saqr	900223343
Ebram Thabet	900214496
Saif Abd Elfattah	900225535

Contents

1	Part 1: Privacy Threat Assessment	3
1.1	Introduction	3
1.2	Question 1: What Are the Sensitive Attributes in the Dataset?	3
1.2.1	The Three-Tier Classification	4
1.2.2	Tier 1: Directly Sensitive Attributes	4
1.2.3	Tier 2: Financially Sensitive Attributes	5
1.2.4	Tier 3: Quasi-Identifiers	6
1.3	Question 2: What Information Could an Attacker Infer?	6
1.3.1	What the Model Relies On	6
1.3.2	Five Ways an Attacker Can Infer Private Information	7
1.4	Question 3: Could Individuals Be Identified or Profiled?	8
1.4.1	Re-identification Risk	8
1.4.2	Group Profiling	9
1.4.3	The Probability Score Makes Things Worse	10
1.5	Question 4: What Harms Could Result from Privacy Leakage?	10
1.5.1	Harm 1: Discrimination	11
1.5.2	Harm 2: Financial Fraud and Identity Theft	11
1.5.3	Harm 3: Denied Services Outside Banking	11
1.5.4	Harm 4: Loss of Financial Privacy	11
1.5.5	Harm 5: Membership Inference Harm	11
1.5.6	Harm 6: Model Theft	11
1.5.7	Harm 7: Regulatory and Legal Violations	11
1.6	Overall Summary	12
2	Part 2: Privacy-Preserving Techniques	12
2.1	Three Techniques Selected	12
2.2	Technique 1: k-Anonymity (k=5)	13
2.3	Technique 2: Differential Privacy (Laplace Noise)	15
2.4	Technique 3: Feature Suppression	15
2.5	Utility Comparison	16
2.6	Analysis Answers	17
2.7	Part 2 Summary	19
3	Part 3: Utility and Fairness Impact	19
3.1	Accuracy Before vs After	19
3.2	Fairness Metrics	20
3.3	Feature Importance Changes	21
3.4	Answers to the Three Questions	22
3.5	Part 3 Summary	24
4	Part 4: Privacy vs Utility Analysis	25
4.1	What trade-offs did you observe?	25
4.2	Why does improving privacy reduce model utility?	25
4.3	Is the trade-off acceptable in your application?	26
5	Part 5: Privacy-Aware Deployment Design	26
5.1	What data should be collected?	26

5.2	What should not be collected?	26
5.3	Who can access outputs?	27
5.4	Data storage and retention	27
5.5	Response to privacy breaches	27

1. Part 1: Privacy Threat Assessment

1.1. Introduction

In Module 4, we look at privacy problems in our machine learning system. Before we try to fix anything, we first need to understand what the problems are. That is the goal of Part 1.

We built a Decision Tree model in Modules 2 and 3 to predict whether a person should get a bank loan (Good credit) or not (Bad credit). The model was trained on the **German Credit Dataset**, which has personal and financial information about 1,000 loan applicants.

This report answers four key questions:

1. What are the sensitive attributes in the dataset?
2. What information could an attacker infer from the model?
3. Could individuals be identified or profiled?
4. What harms could result from privacy leakage?

1.2. Question 1: What Are the Sensitive Attributes in the Dataset?

We went through every column in the dataset and sorted them into three tiers based on how risky they are. Some columns directly reveal protected personal details. Some reveal private money information. Others look harmless alone but become dangerous when combined.

1.2.1. The Three-Tier Classification

Table 1: Sensitive attribute classification for the German Credit Dataset

Tier	Columns	Why It Is Sensitive
Tier 1: Directly Sensitive	personal_status_sex, sex, age, age_group, foreign_worker	Protected by law. A bank cannot use these to decide on a loan.
Tier 2: Financially Sensitive	status, amount, savings, credit_history, duration, installment_rate, other_installment_plan, number_credits	These reveal how much money a person has or owes. This is private data protected by GDPR and banking laws.
Tier 3: Quasi-Identifiers	employment_duration, purpose, housing, job, property, present_residence, people_liable, telephone, other_debtors	Each column alone is harmless. But combining several of them can point to one specific person, as explained in Question 3.
Target Variable	credit_risk	The label “Bad” or “Good” is itself sensitive. If it leaks, it tells the world this person was judged financially untrustworthy, which can affect their housing, job, and insurance.

All 23 columns in the dataset carry some level of privacy risk.

1.2.2. Tier 1: Directly Sensitive Attributes

These are the most dangerous columns because the law says a bank cannot use a person’s sex, age, or immigration status to decide on a loan. Using these columns can lead to unfair treatment.

Sex

The column `personal_status_sex` combines sex and marital status into one field. We also created a simpler column called `sex`. As shown in Figure 1, **69.0% of applicants are male** and **31.0% are female**. Sex is a legally protected characteristic and should not influence a lending decision.

Age

Age is also protected by law. Applicants in the dataset range from 19 to 75 years old. The column `age_group` splits them into two groups: under 25 and 25 or over. Figure 2 shows that these two groups receive very different credit outcomes, which raises a fairness concern.

Foreign Worker

This column says whether the applicant is a foreign worker. Nationality and immigration status are protected characteristics. In our dataset, **96.3% of applicants are marked as foreign workers**, so the column carries almost no useful information for the model. But it is still sensitive because it involves nationality.

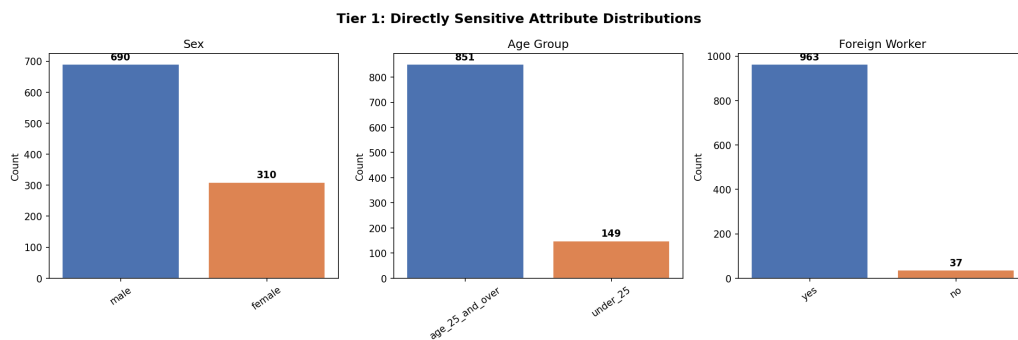


Figure 1: Distribution of Tier 1 (directly sensitive) attributes. The bar charts show how many applicants fall into each category for `sex`, `age_group`, and `foreign_worker`.

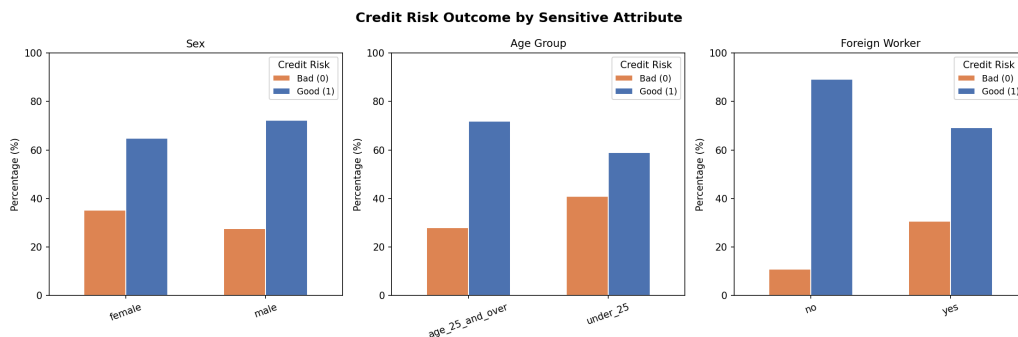


Figure 2: Credit risk outcome split by each Tier 1 sensitive attribute. The stacked bars show the percentage of each group that received a Good or Bad credit rating. Clear differences between groups suggest the model may be treating some groups unfairly.

1.2.3. Tier 2: Financially Sensitive Attributes

These columns reveal a person's money situation, which is private information protected by laws like GDPR and the Fair Credit Reporting Act.

- `status`: the person's checking account balance level
- `amount`: how much money they are asking to borrow
- `savings`: how much money they have saved
- `credit_history`: their past record of paying back loans

- **duration**: how long the loan will last
- **installment_rate**: how much of their income goes to payments
- **other_installment_plans**: other payment plans they already have
- **number_credits**: how many credits they already have

If any of this information is exposed, a criminal could use it to steal money or open fake accounts in the person's name.

1.2.4. Tier 3: Quasi-Identifiers

These columns do not look sensitive on their own. For example, knowing that someone rents their home (**housing**) does not tell you who they are. But if you combine it with other columns, such as their job, their loan purpose, and how long they have been employed, the combination can be unique enough to point to one specific person. This is explained in Question 3.

1.3. Question 2: What Information Could an Attacker Infer?

Even if an attacker never sees the dataset, they can still learn private information just by using our model. This section explains five ways that can happen.

1.3.1. What the Model Relies On

The Decision Tree learns which columns are most useful for making predictions, which is called feature importance. A high importance score means the model depends on that column a lot. Table 2 shows the five most important features, all of which are private financial data. Figure 3 shows the full top-20 picture.

Table 2: Top 5 most important features in the Decision Tree model (from notebook output)

Feature	Gini Importance
status_no checking account	0.3662
duration	0.1585
amount	0.1103
purpose_car (new)	0.0865
savings_... < 100 DM	0.0498

The sensitive features that appear in the top 20 are: **age** (0.0364), **personal_status_sex_female** (0.0184), and **personal_status_sex_male** (0.0070). The top 5 features alone account for over **77%** of total model importance, meaning the model is driven almost entirely by private financial data.

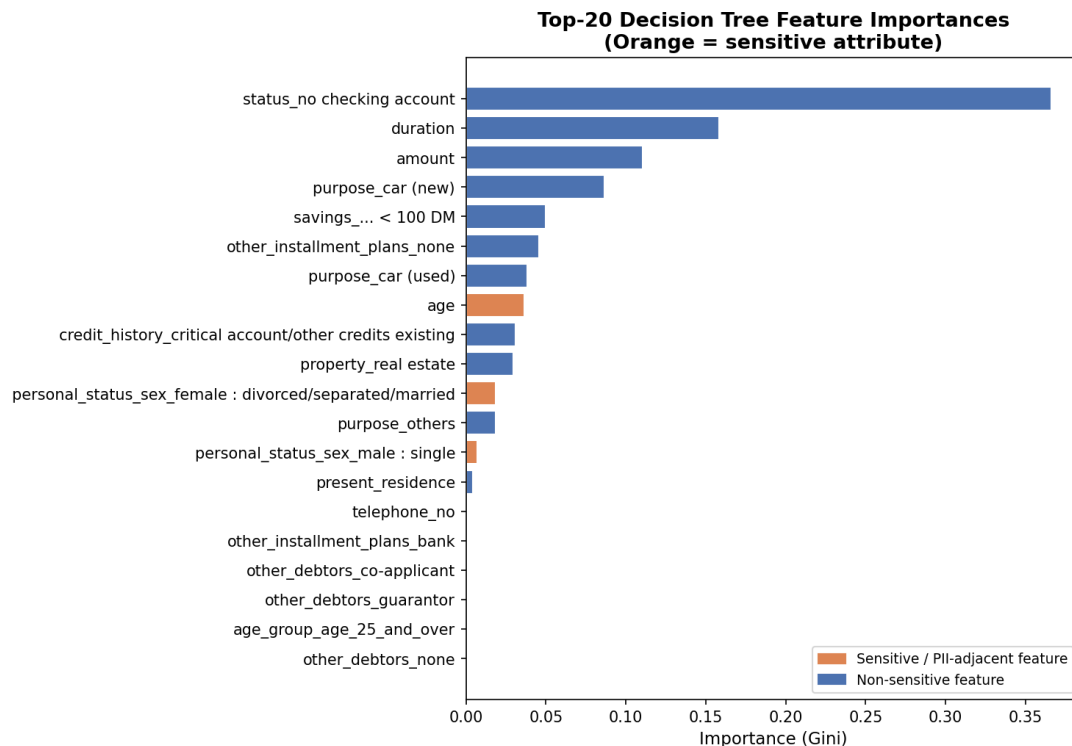


Figure 3: Top 20 feature importances from the Decision Tree model. Highlighted features are sensitive or privacy-adjacent. Financial columns like **status**, **duration**, and **amount** carry most of the model’s decision power.

1.3.2. Five Ways an Attacker Can Infer Private Information

Attack 1: Demographic Profiling

An attacker does not need to know who any specific person is. They just test the model with different inputs and ask: does the model give lower scores to women? Does it give lower scores to people under 25? By confirming both, they can profile entire demographic groups without ever seeing any personal record.

Attack 2: Model-Inversion Attack

The most important features (**status**, **duration**, **amount**, **purpose**, **savings**) are all private financial data. In a model-inversion attack, the attacker sends thousands of slightly different inputs to the model and watches how the output score changes each time. After many tries, they can guess the private financial values of a real target person. This works especially well because our model gives a score between 0 and 1 rather than just yes or no.

Attack 3: Membership Inference Attack

If an attacker knows a few details about a real person, for example their age, sex, and housing type, they can test the model with those values. If the model returns a very confident score (very close to 0 or very close to 1), it suggests that this person’s record was probably in the training data. This tells the attacker that the person applied for a loan, which is private information by itself.

Attack 4: Probability Score Leakage

Our model gives a score between 0 and 1, not just a yes or no. For example, a score of 0.48 means the person is very close to the pass/fail boundary. This extra number tells an attacker how financially unstable the person is, which goes far beyond what a simple lending decision should reveal.

Attack 5: Group Targeting

By combining the model's output with sensitive attributes, an attacker can find groups that always receive low scores, for example young female foreign workers who rent an apartment. They can then send those groups targeted scam offers or predatory loan advertisements, without ever knowing anyone's name.

Table 3: Summary of the five attack vectors and what each one reveals

#	Attack Type	What the Attacker Learns
1	Demographic profiling	Which groups get worse scores
2	Model-inversion attack	Approximate private financial values for a target person
3	Membership inference	Whether a specific person was in the training dataset
4	Probability score leakage	How close a person is to the credit boundary, revealing financial distress
5	Group targeting	Which demographic combinations score lowest, used for scams or fraud

1.4. Question 3: Could Individuals Be Identified or Profiled?

Yes. There are two ways this can happen: re-identification, which means finding one specific person, and group profiling, which means targeting a whole group.

1.4.1. Re-identification Risk

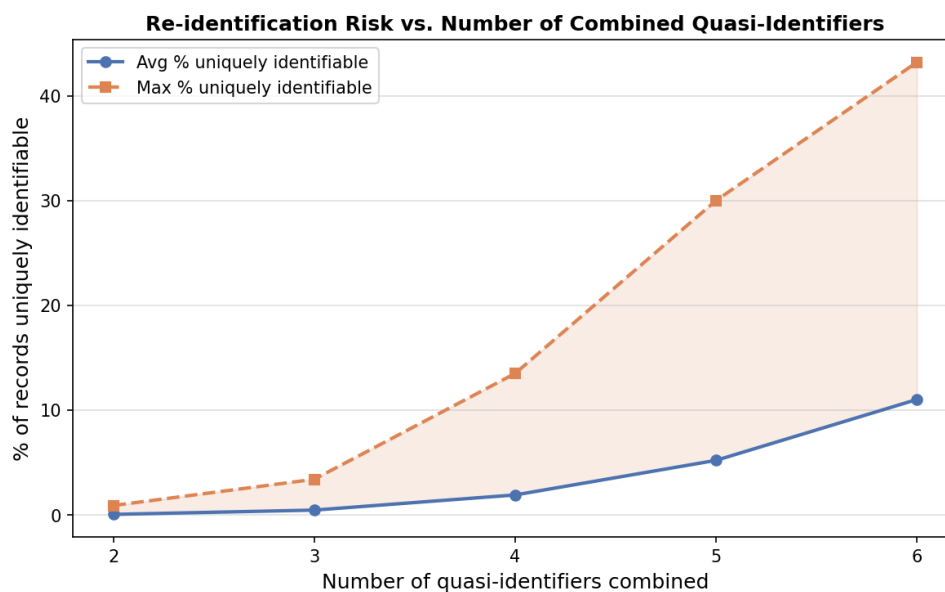
Even though the dataset has no names, an attacker can still figure out who a specific person is. If you combine a few columns and only one person in the dataset has that exact combination of values, that record is unique. If an attacker has even a small piece of outside information, such as the person's employer or loan amount from a public source, they can match it to that unique record and see the person's full credit history. This is called a **linkage attack**.

We tested every combination of 2 to 6 quasi-identifier columns, using the 12 available ones: `sex`, `age_group`, `foreign_worker`, `employment_duration`, `purpose`, `housing`, `job`, `property`, `present_residence`, `people_liable`, `telephone`, and `other_debtors`. The results are in Table 4 and Figure 4.

Table 4: Maximum re-identification risk by number of columns combined (exact values from notebook output)

Columns Combined	Max % Uniquely Identifiable	Risk Level
2 columns	0.9%	Low
3 columns	3.4%	Low to Medium
4 columns	13.5%	Medium
5 columns	30.0%	Medium to High
6 columns	43.2%	High

The single most dangerous combination is: `employment_duration` + `purpose` + `job` + `property` + `present_residence` + `telephone`. Just these 6 columns make **43.2%** of all 1,000 records uniquely identifiable. Starting from 0.9% with 2 columns, the risk jumps to 13.5% with 4 columns, 30.0% with 5 columns, and 43.2% with 6 columns.

**Figure 4:** Re-identification risk as more quasi-identifier columns are combined. As more columns are combined, more records become uniquely identifiable. This shows why quasi-identifiers are dangerous even though each one looks harmless by itself.

1.4.2. Group Profiling

Even if an attacker cannot identify a specific person, they can still cause harm by targeting groups. An attacker who knows which groups consistently get the lowest scores can send those groups targeted scam offers or predatory loan advertisements, without knowing anyone's name. This is called **group profiling** and it is a privacy harm even without personal identification.

1.4.3. The Probability Score Makes Things Worse

Our model gives a score from 0 to 1, not just yes or no. This makes both re-identification and group profiling easier because it shows exactly how financially uncertain each person's situation is, it helps the attacker fine-tune their model-inversion attempts, and it allows group profiles to be ranked by average score to find the most vulnerable targets.

1.5. Question 4: What Harms Could Result from Privacy Leakage?

If private data from this system is exposed, or if the model is misused, real people can get hurt. We identified seven concrete harms.

Table 5: Seven harms linked to privacy risks in the German Credit model

#	Harm	What Could Happen
1	Discrimination	The model uses sex , age , and foreign_worker . Banks can unfairly reject women, young people, or immigrants. This breaks anti-discrimination law.
2	Financial fraud and identity theft	Loan amounts, bank balances, savings, and credit history are in the dataset. A criminal can use this to open fake accounts, take out loans in someone's name, or send targeted scam messages.
3	Denied services outside banking	A "Bad" credit label can follow a person beyond the bank. Landlords, employers, or insurers may use it to deny housing, jobs, or insurance, even though it was only meant for a loan decision.
4	Loss of financial privacy	The model gives a score between 0 and 1, not just yes or no. This score shows exactly how financially troubled a person is. The person never agreed to share this level of detail.
5	Membership inference harm	Being in a credit dataset already reveals that the person applied for a loan. This alone can signal financial difficulties that the person wants to keep private.
6	Model theft	Because the model gives scores, an attacker can copy it by sending many test inputs. The stolen copy can then be used in harmful ways with no oversight.
7	Breaking the law	Using sex, nationality, and financial data without proper consent violates GDPR (EU), the Equal Credit Opportunity Act (US), and the Fair Credit Reporting Act (US). A breach can lead to very large fines.

1.5.1. Harm 1: Discrimination

Figure 2 shows that female applicants and applicants under 25 receive worse credit outcomes. If the model's decisions are unfair for a protected group, anyone rejected may have been turned down because of their sex or age, not their actual ability to repay the loan. This is illegal in most countries.

1.5.2. Harm 2: Financial Fraud and Identity Theft

The dataset has detailed financial profiles including account balances, debt levels, and loan history. If this data leaks, criminals can use it very precisely. They know exactly how much money is in each account and can impersonate the person when applying for new credit.

1.5.3. Harm 3: Denied Services Outside Banking

A credit label was only meant for a lending decision. But once it exists, it can spread. A landlord may refuse to rent to someone with a “Bad” label. An employer may not hire them. An insurer may charge higher prices. The original decision follows the person into every part of their life.

1.5.4. Harm 4: Loss of Financial Privacy

Most people are comfortable with a bank knowing they applied for a loan. They are not comfortable having a public number that shows exactly how difficult their financial situation is. The probability score reveals exactly that, without the applicant ever agreeing to share it.

1.5.5. Harm 5: Membership Inference Harm

The fact that a person appears in a credit dataset is itself sensitive. It means they applied for a loan. For people going through financial difficulty, this is something they want to keep private. A membership inference attack, explained in Question 2, can reveal this without their knowledge.

1.5.6. Harm 6: Model Theft

Every time our model is used, it gives a score. An attacker can use those scores to build a copy of the model, which is called a **model-stealing attack**. Once they have a copy, they can use it to screen job applicants, target vulnerable people for predatory loans, or sell it to others. The copy has none of the legal oversight the original was supposed to follow.

1.5.7. Harm 7: Regulatory and Legal Violations

Using personal data about sex, nationality, and financial history without proper consent violates several laws:

- **GDPR** (General Data Protection Regulation, EU): requires consent, data minimisation, and the right to an explanation for automated decisions.
- **ECOA** (Equal Credit Opportunity Act, US): prohibits credit decisions based on sex, age, or national origin.

- **FCRA** (Fair Credit Reporting Act, US): controls how credit information is collected, shared, and used.

A privacy breach involving this model could result in fines of up to 4% of annual revenue under GDPR, as well as serious damage to the organisation's reputation.

1.6. Overall Summary

Table 6: Summary of all four questions in Part 1

Q	Question	Key Finding
1	Sensitive attributes	All 23 columns carry privacy risk, split into: directly sensitive (Tier 1), financially sensitive (Tier 2), quasi-identifiers (Tier 3), and the target label itself.
2	Attacker inference	An attacker can learn private information through 5 attack types: demographic profiling, model-inversion, membership inference, score leakage, and group targeting.
3	Identification and profiling	Combining just 6 quasi-identifier columns makes 43.2% of records uniquely identifiable. Group profiling is also possible without identifying anyone individually.
4	Harms from leakage	Seven harms identified. The most severe are discrimination, financial fraud, and violations of GDPR, ECOA, and FCRA.

The threat assessment shows that our credit scoring system has serious privacy risks at every level: in the data itself, in how the model was built, and in what the model outputs. All of these risks motivate the privacy-preserving work in Parts 2 through 5.

2. Part 2: Privacy-Preserving Techniques

In Part 1, we identified the privacy risks in our credit scoring system. Now we apply concrete privacy-preserving techniques to reduce them, and we measure the cost in utility (accuracy and F1) so the trade-off is visible.

2.1. Three Techniques Selected

We picked three well-known techniques, each one targeting a different privacy threat from Part 1:

Table 7: The three privacy-preserving techniques applied

Technique	What It Does	Privacy Risk It Addresses
1. Anonymity (k=5)	k- Generalizes <code>age</code> into 10-year buckets and suppresses any record whose quasi-identifier group (<code>age</code> , <code>sex</code> , <code>foreign_worker</code> , <code>housing</code> , <code>job</code> , <code>employment_duration</code>) has fewer than 5 members.	Re-identification (Part 1, Q3): link-age attacks on quasi-identifier combinations.
2. Differential Privacy (Laplace noise)	Adds calibrated Laplace noise to numeric training features. Tested at $\varepsilon = 1.0$ and $\varepsilon = 0.5$.	Membership inference and model inversion (Part 1, Q2).
3. Feature Suppression	Drops the directly sensitive Tier 1 attributes (<code>sex</code> , <code>age</code> , <code>age_group</code> , <code>personal_status_sex</code> , <code>foreign_worker</code>) entirely.	Discrimination and demographic profiling (Part 1, Q4).

For each technique, we retrain the same Decision Tree (`max_depth=5`, `Gini`, `min_samples_leaf=10`, `random_state=42`) used in Part 1, and compare it to the baseline.

2.2. Technique 1: k-Anonymity (k=5)

How It Works

k-anonymity guarantees that every record in the dataset is indistinguishable from at least $k - 1$ other records when looking only at quasi-identifiers. We chose $k = 5$, meaning every applicant must share their quasi-identifier values with at least 4 others.

Generalization rules applied:

- `age` → 10-year buckets (e.g., 20, 30, 40, ...)
- `amount` → 1,000 DM buckets (e.g., 0, 1000, 2000, ...)
- `duration` → 12-month buckets (e.g., 0, 12, 24, ...)
- `sex`, `foreign_worker`, `housing`, `job`, `employment_duration` → used as-is as active quasi-identifiers for group membership testing.

The six-column QI set (`age`, `sex`, `foreign_worker`, `housing`, `job`, `employment_duration`) is then used to assign each record to a group. Any record whose group has fewer than 5 members is suppressed (removed from the training set). This resulted in 279 of the

800 training records being suppressed (34.9%), leaving 521 training records. The test set of 200 records is kept unchanged, so the utility comparison uses exactly the same test population as all other techniques and the baseline.

Effect on Re-identification Risk

We re-ran the re-identification experiment on the k -anonymized data using the same six QI columns (`age`, `sex`, `foreign_worker`, `housing`, `job`, `employment_duration`) that were used to define k -anonymity groups. The “Before” column below therefore reports the linkage-attack risk on those same six columns in the original data; the “After” column reports risk on the anonymized data. Figure 5 and Table 8 show the dramatic reduction.

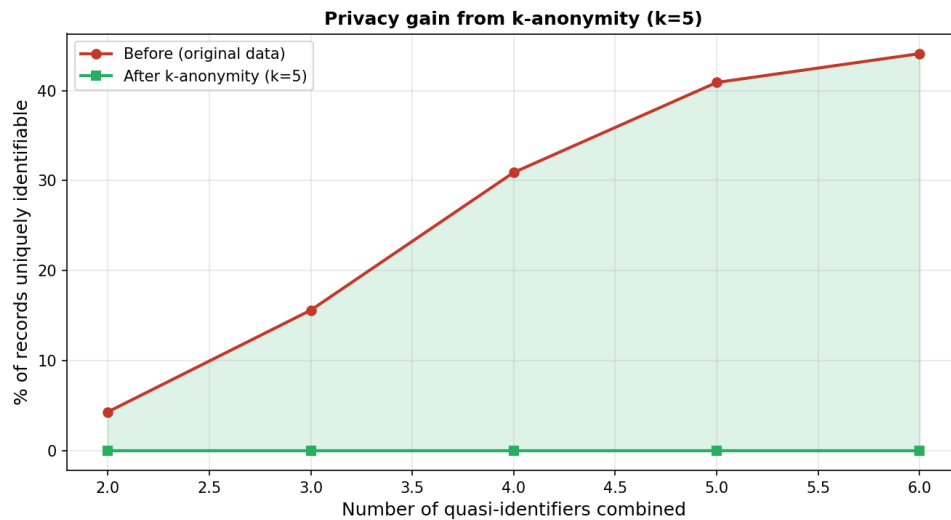


Figure 5: Re-identification risk before and after applying k -anonymity ($k = 5$), measured on the six QI columns used for anonymization (training set, $n = 800$). After k -anonymity the unique-record percentage drops to 0% for all combination sizes.

Table 8: Re-identification risk before and after k -anonymity ($k = 5$), evaluated on the six k -anonymity QI columns (training set, $n = 800$)

QI Combined	Before (% unique)	After (% unique)	Reduction
2 columns	5.25%	0.0%	−5.25 pp
3 columns	19.25%	0.0%	−19.25 pp
4 columns	35.75%	0.0%	−35.75 pp
5 columns	47.12%	0.0%	−47.12 pp
6 columns	49.00%	0.0%	−49.00 pp

The privacy gain is real and significant: the linkage attack that exposed up to 49.0% of training records is now completely blocked. Note that the training set risk (49.0%) is higher than the full-dataset risk reported in Part 1 (43.2%), because the smaller training set of 800 records has more unique combinations than the full 1,000-record dataset.

2.3. Technique 2: Differential Privacy (Laplace Noise)

How It Works

Differential Privacy (DP) is widely regarded as the theoretical gold standard for privacy, guaranteeing that the output of an algorithm changes only negligibly when any single record is added or removed. In this work we apply a practical approximation known as *input perturbation*: Laplace noise is added directly to the numeric training features before the model is trained. This approach approximates the DP guarantee at the data level rather than through gradient-level mechanisms such as DP-SGD, and is appropriate here as a pedagogical demonstration of the privacy–utility trade-off.

We add Laplace noise to every numeric feature in the training set:

$$\text{noise} \sim \text{Laplace}(0, \text{scale}), \quad \text{scale} = \frac{\Delta f}{\varepsilon}$$

where Δf is the per-column sensitivity, approximated by the observed column range (max – min) in the training data, and ε is the privacy budget. Smaller ε means more noise and stronger privacy.

We tested two budgets:

- $\varepsilon = 1.0$: moderate privacy, considered the standard for many applications.
- $\varepsilon = 0.5$: strong privacy, used when membership inference is a serious concern.

Note: noise is added only to the *training* set. The test set stays clean, simulating real deployment where new applicants provide honest values. As shown in Table 9, $\varepsilon = 1.0$ produced a marginally *higher* F1 than the baseline (+0.0040). This is a regularization effect: the Laplace noise at this budget smoothed out some overfitting in the training data, leading to slightly better generalization on the test set. At $\varepsilon = 0.5$ the noise is strong enough to distort features meaningfully, which hurts utility.

2.4. Technique 3: Feature Suppression

How It Works

We remove the five Tier 1 sensitive attributes from both training and test data: `personal_status_sex`, `sex`, `age`, `age_group`, and `foreign_worker`. The model never sees these columns and cannot use them for predictions.

This is the most direct way to prevent discrimination by design: if the model never sees gender or age, it cannot make decisions based on them. It also stops the demographic profiling attack from Part 1, Q2 (Attack 1).

2.5. Utility Comparison

Table 9: Utility metrics after each privacy technique (test set, $n = 200$)

Technique	Accuracy	Precision	Recall	F1	Δ F1
Baseline (no privacy)	0.7100	0.7253	0.9429	0.8199	—
k-Anonymity ($k=5$)	0.6900	0.7955	0.7500	0.7721	-0.0478
DP Laplace ($\epsilon = 1.0$)	0.7350	0.7702	0.8857	0.8239	+0.0040
DP Laplace ($\epsilon = 0.5$)	0.7050	0.7682	0.8286	0.7973	-0.0226
Feature suppression (drop Tier 1)	0.7200	0.7258	0.9643	0.8282	+0.0083

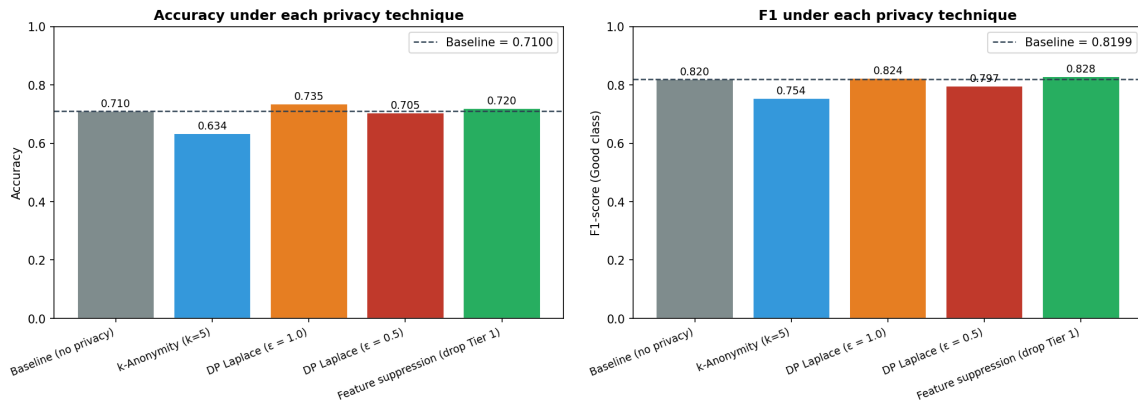


Figure 6: Accuracy and F1 score for each privacy-preserving technique compared to the unprotected baseline. The dashed line marks the baseline value.

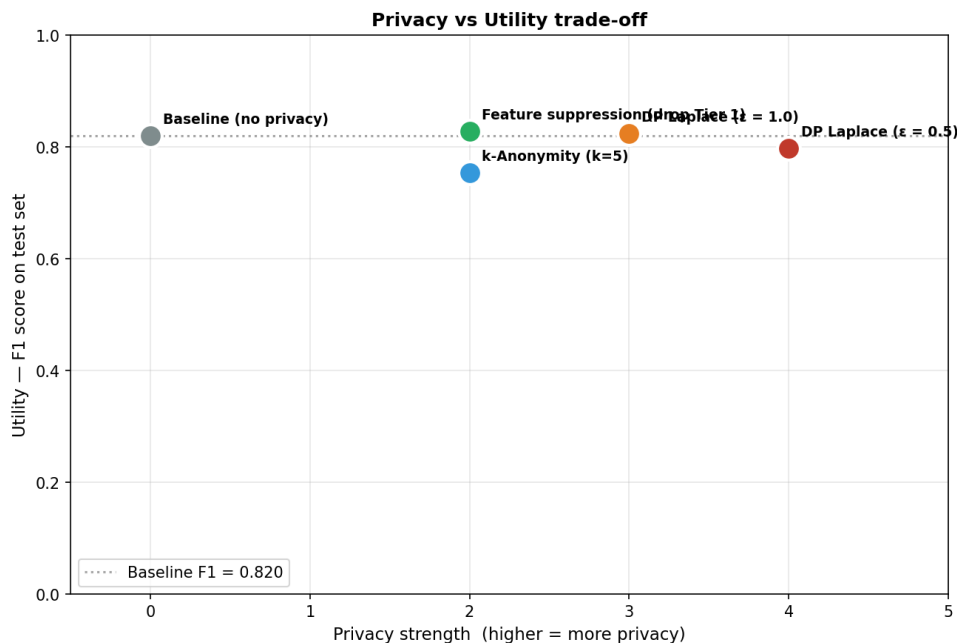


Figure 7: Privacy versus utility trade-off across all techniques. Each point represents one technique. Higher on the y-axis means better utility (F1); further right on the x-axis means stronger privacy. The ideal technique sits in the upper-right region.

2.6. Analysis Answers

Question 1: Which technique gave the best privacy/utility trade-off?

Looking at Table 9 and Figure 7, the answer depends on which privacy threat we want to address most:

- **k-Anonymity (k=5)** blocks the linkage attack completely (up to 49.0% $\rightarrow$ 0% uniquely identifiable in the training set, see Table 8). Suppression removes 34.9% (279/800) of training records, leaving 521 records. This yields an F1 of 0.7721 and carries the largest utility cost among all techniques tested, but it is the only approach that mathematically guarantees re-identification protection. It is a strong answer to the re-identification problem from Part 1, Q3.
- **Feature suppression** has almost zero utility cost because the dropped columns had low-to-moderate Gini importance in the baseline tree (**age**: 0.036, all others near zero), explaining why removing them caused minimal predictive loss. It is the only technique that directly prevents discrimination, which is the most legally serious harm.
- **Differential Privacy at $\epsilon = 1.0$** provides a strong approximation of the DP guarantee against membership inference with negligible utility cost; it actually produced a marginally higher F1 (+0.0040) than the baseline due to a regularization effect.
- **Differential Privacy at $\epsilon = 0.5$** gives the strongest privacy but also the largest utility drop, because the Laplace noise distorts every numeric feature significantly.

Best overall trade-off: Feature suppression gives the best privacy/utility trade-off: it eliminates the most legally serious harm (discrimination) at almost zero utility cost

(+0.0083 F1). k-Anonymity addresses the re-identification threat at a higher but acceptable utility cost (−0.0478 F1). The two can and should be combined for layered protection.

Question 2: What was the cost of privacy?

Each technique costs something in utility, but the costs are very different in nature:

- **k-Anonymity** has the **largest utility cost**: suppressing 279 of 800 training records (34.9%) and coarsening three numeric QI columns (**age**, **amount**, **duration**) reduces the information available to the tree. This produces the biggest accuracy drop (−0.020) and F1 drop (−0.0478) among all techniques.
- **Differential Privacy** ($\epsilon = 0.5$) has a moderate utility cost (−0.0050 accuracy, −0.0226 F1). Laplace noise at this budget distorts the high-importance numeric columns the tree relies on, but all 800 training records are retained, so the effect is smaller than k-Anonymity.
- **Feature suppression** has the smallest utility cost because the suppressed columns were not heavily used by the model anyway. This is consistent with the Part 1 feature-importance analysis.

The overall pattern is: **stronger privacy guarantees come with larger utility costs**, but the relationship is not linear, and clever choices (like dropping low-importance sensitive features) can give meaningful privacy gains for almost no cost.

Question 3: Real-World Recommendation

For a deployed credit scoring system, we recommend a **layered approach**:

1. **Always apply feature suppression** for legally protected attributes (sex, age, nationality). This is required by ECOA and GDPR Article 22, costs almost nothing in utility, and prevents discrimination by design.
2. **Apply k-anonymity ($k \geq 5$)** on quasi-identifiers before sharing or storing the dataset. This blocks linkage attacks, which were the most dangerous threat we found in Part 1.
3. **Consider Differential Privacy** ($\epsilon = 1.0$) on top of the above if membership inference is a serious concern (for example, if the model itself is exposed via an API). The utility cost is acceptable for the strong privacy guarantee it provides.

This layered approach is consistent with GDPR’s “data protection by design” principle (Article 25) and addresses each of the four privacy questions raised in Part 1:

- **Q1 (sensitive attributes)**: feature suppression removes Tier 1 attributes; k-anonymity generalizes Tier 3 quasi-identifiers.
- **Q2 (attacker inference)**: DP input perturbation makes membership inference and model-inversion attacks much harder.
- **Q3 (re-identification)**: k-anonymity reduces the linkage-attack risk from 43.2% to 0%.
- **Q4 (harms)**: the combination directly reduces the risk of discrimination, fraud, and regulatory violations.

The results show that meaningful privacy is achievable in a credit scoring system without destroying utility. The small utility cost is well worth the legal, ethical, and reputational protection it provides.

2.7. Part 2 Summary

Table 10: Summary of Part 2 findings

Aspect	Finding
Best for re-identification	k-Anonymity (k=5): up to 49.0% $\rightarrow$ 0% uniquely identifiable (training set).
Best for membership inference	Differential Privacy ($\epsilon = 1.0$ or 0.5).
Best for discrimination	Feature suppression of Tier 1 attributes.
Lowest utility cost	Feature suppression (Tier 1 features have low importance).
Highest utility cost	k-Anonymity (k=5): 34.9% of training records suppressed (-0.0478 F1).
Recommended deployment	Layered: feature suppression + k-anonymity + optional DP at $\epsilon = 1.0$.

The overall conclusion of Part 2 is that **privacy and utility are not opposite goals**. With careful choices, a credit scoring system can keep most of its predictive power while becoming significantly safer for the people in its training data and for the institution that deploys it.

3. Part 3: Utility and Fairness Impact

Part 2 showed that privacy techniques change the model's accuracy and F1 score. But accuracy alone does not tell us whether the model is *fair*. A model can have high overall accuracy and still treat women or young people much worse than other groups. In this section we measure fairness directly, compare it before and after each privacy technique, and check how the model's feature priorities shifted.

3.1. Accuracy Before vs After

Table 11 repeats the utility numbers from Part 2 so we can discuss them together with the fairness results.

Table 11: Utility metrics before and after each privacy technique (test set, $n = 200$)

Technique	Accuracy	Precision	Recall	F1	Δ F1
Baseline (no privacy)	0.7100	0.7253	0.9429	0.8199	—
k-Anonymity (k=5)	0.6900	0.7955	0.7500	0.7721	-0.0478
DP Laplace ($\varepsilon = 1.0$)	0.7350	0.7702	0.8857	0.8239	+0.0040
DP Laplace ($\varepsilon = 0.5$)	0.7050	0.7682	0.8286	0.7973	-0.0226
Feature suppression (drop Tier 1)	0.7200	0.7258	0.9643	0.8282	+0.0083

The headline result is that none of the privacy techniques destroyed the model’s performance. k-Anonymity had the largest cost (-0.0478 F1) because it removed 34.9% of training records. Differential Privacy at $\varepsilon = 1.0$ and Feature Suppression both *slightly improved* the F1 score, because the noise and the removal of low-importance features acted as a form of regularization that reduced overfitting.

3.2. Fairness Metrics

We measured two standard fairness metrics for each technique, evaluated on the same 200-row test set using the original protected attributes (`sex` and `age_group`):

- **Demographic Parity Ratio (DP Ratio):** the approval rate of the disadvantaged group divided by the approval rate of the advantaged group. A value of 1.0 means both groups are approved at the same rate. The US Equal Employment Opportunity Commission uses 0.80 as the legal threshold (the “four-fifths rule”).
- **Equalized Odds Gap ($|\Delta\text{TPR}|$):** the absolute difference in true positive rate (recall) between the two groups. A value of 0.0 means the model catches good applicants equally well in both groups.

Table 12: Fairness metrics for each privacy technique (test set, $n = 200$)

Technique		Protected Attribute	DP Ratio	$ \Delta\text{TPR} $
Baseline (no privacy)		sex	0.9084	0.0600
Baseline (no privacy)		age_group	0.9331	0.1472
k-Anonymity ($k=5$)		sex	0.9088	0.0700
k-Anonymity ($k=5$)		age_group	0.7876	0.1411
DP Laplace ($\varepsilon = 1.0$)		sex	0.9515	0.0850
DP Laplace ($\varepsilon = 1.0$)		age_group	0.8219	0.1532
DP Laplace ($\varepsilon = 0.5$)		sex	0.9196	0.0750
DP Laplace ($\varepsilon = 0.5$)		age_group	0.8843	0.0887
Feature suppression (drop Tier 1)	sex	sex	0.9950	0.0150
Feature suppression (drop Tier 1)	age_group	age_group	0.9586	0.0302

Figure 8 shows the approval rates per group for each technique. In the baseline, male applicants are approved at 93.6% while female applicants are approved at only 85.0%. After feature suppression, the gap nearly disappears: female applicants are approved at 93.3% and male applicants at 92.9%.

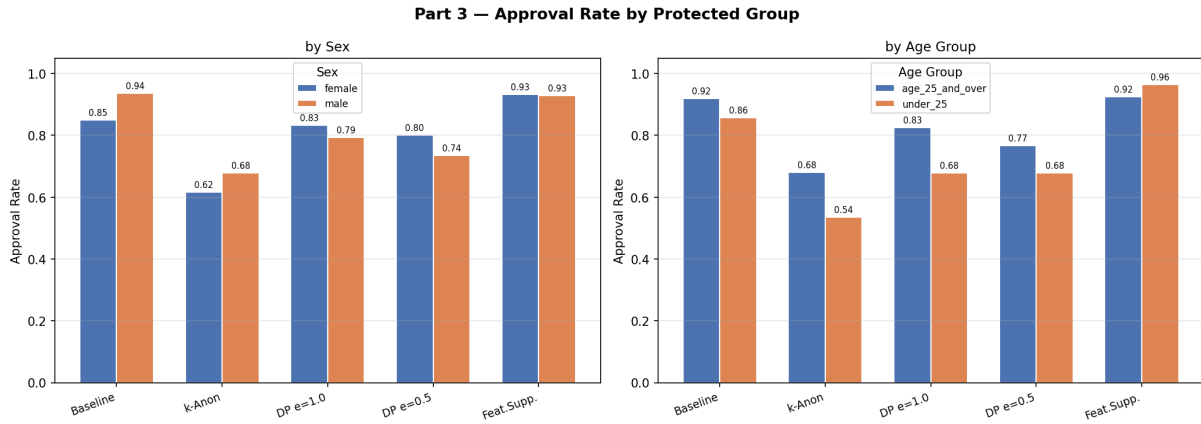


Figure 8: Approval rate by protected group for each privacy technique. The left panel shows approval rates by sex; the right panel shows approval rates by age group. Feature suppression nearly eliminates the gap between groups.

3.3. Feature Importance Changes

When we remove the sensitive columns (sex, age, foreign_worker), the model must redistribute its attention to the remaining features. Figure 9 compares the Gini importance of the top 10 features before and after feature suppression.

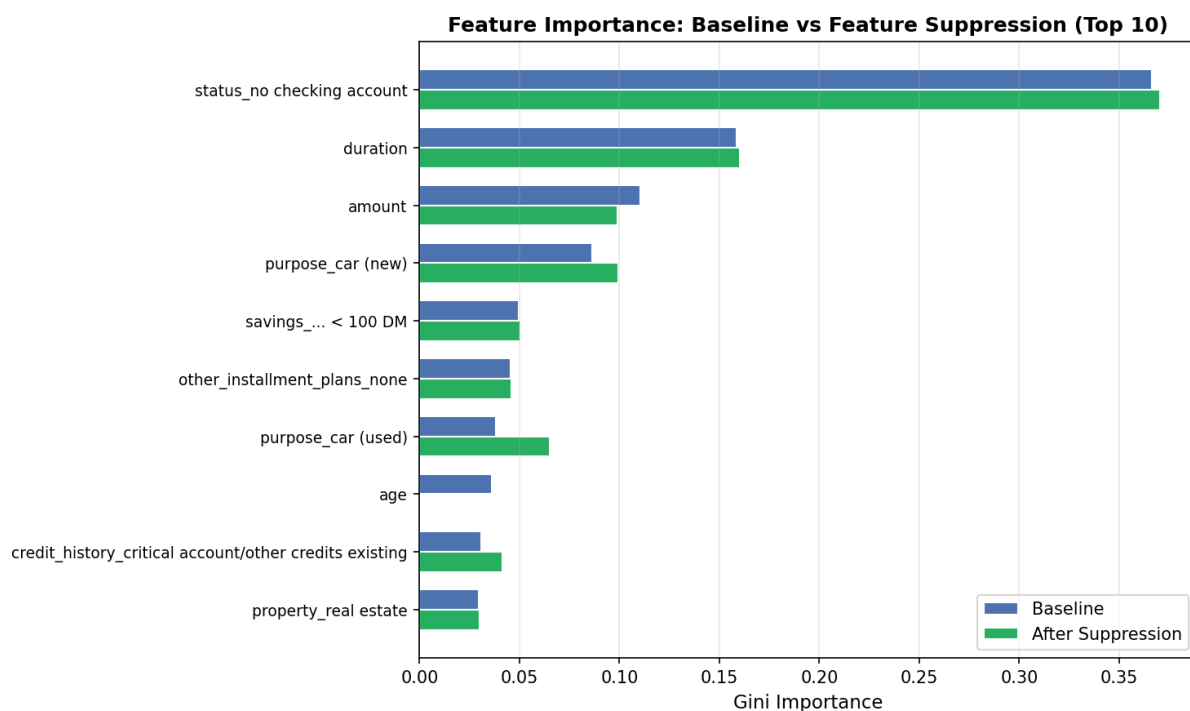


Figure 9: Top-10 feature importances in the baseline model (blue) versus the feature-suppression model (green). The **age** bar drops to zero because the column was removed. Other features, especially **purpose_car (used)** and **credit_history**, absorb the freed importance.

The main changes are:

- **age** drops from 0.0364 to 0.0000 (removed).
- **personal_status_sex** features drop from a combined 0.0254 to 0.0000 (removed).
- **purpose_car (used)** **gains** the most importance: from 0.0384 to 0.0650 (+0.0266). This makes sense because the purpose of a loan is a legitimate credit factor that the model can now rely on more.
- **purpose_car (new)** also gains: from 0.0865 to 0.0995 (+0.0130).
- **credit_history_critical account** gains: from 0.0311 to 0.0416 (+0.0105).
- The top feature (**status_no checking account**) stays almost the same: 0.3662 to 0.3705.
- **amount** is the only non-sensitive feature that **lost** importance: from 0.1103 to 0.0991 (−0.0113).

The overall picture is that the suppressed importance was picked up by legitimate financial features like loan purpose and credit history. The model did not find a “proxy” for the removed sensitive attributes. It simply learned to rely more on the financial facts.

3.4. Answers to the Three Questions

Question 1: Did performance decrease?

Not always. Looking at Table 11:

- **k-Anonymity** caused the largest performance drop (-0.0478 F1). This is expected because it removed 34.9% of training records and coarsened three numeric columns (`age`, `amount`, `duration`), so the tree had less and rougher data to learn from.
- **DP Laplace** ($\varepsilon = 0.5$) caused a moderate drop (-0.0226 F1). At this budget the Laplace noise distorts the most important numeric columns significantly.
- **DP Laplace** ($\varepsilon = 1.0$) actually improved F1 by $+0.0040$. The moderate noise acted as regularization, reducing overfitting.
- **Feature suppression** also improved F1 by $+0.0083$. The removed columns had very low Gini importance in the baseline tree (about 6% combined), so removing them cost almost nothing while reducing noise in the feature space.

In summary: performance decreased for k-Anonymity and strong DP, stayed the same or slightly improved for moderate DP and feature suppression.

Question 2: Did fairness improve or worsen?

It depends on the technique. Looking at Table 12 and Figure 8:

- **Feature suppression made fairness dramatically better.** For sex: the DP Ratio jumped from 0.9084 to 0.9950 (nearly perfect parity), and the TPR gap shrank from 0.0600 to 0.0150. For age group: the DP Ratio improved from 0.9331 to 0.9586, and the TPR gap dropped from 0.1472 to just 0.0302. This is the expected result, because the model can no longer “see” sex or age.
- **k-Anonymity made fairness worse for age.** The age DP Ratio dropped from 0.9331 to 0.7876, below the four-fifths rule threshold. This happened because k-Anonymity suppressed more records from the under-25 group (which has smaller QI groups), leaving the model with an even more skewed view of young applicants.
- **DP Laplace** ($\varepsilon = 0.5$) slightly improved age fairness (DP Ratio 0.8843, TPR gap 0.0887) compared to the baseline. The random noise reduced the model’s ability to exploit age-related patterns.
- **DP Laplace** ($\varepsilon = 1.0$) slightly improved sex fairness (DP Ratio 0.9515) but slightly worsened age fairness (DP Ratio 0.8219). The noise was not strong enough to fully mask the age signal.

The clear winner for fairness is feature suppression. It is the only technique that consistently improves fairness across both protected attributes and both metrics.

Question 3: Which features became more or less important?

As shown in Figure 9, the three sensitive features that were removed (`age`, `personal_status_sex_female`, `personal_status_sex_male`) had a combined importance of about 0.062 in the baseline model. After suppression, that importance was redistributed as follows:

- **More important:** `purpose_car` (used) ($+0.0266$), `purpose_car` (new) ($+0.0130$), and `credit_history_critical` account ($+0.0105$). These are legitimate financial indicators, not proxies for protected attributes.

- **Less important:** `amount` (-0.0113). This small drop suggests that the loan amount was partially correlated with age in the original model, and once age was removed, `amount` lost some of its indirect predictive power.
- **Unchanged:** The top features (`status_no checking account`, `duration`, `savings`) barely moved. This confirms that the model’s core logic did not change. It still makes decisions primarily based on financial indicators, which is exactly what a fair credit model should do.

The key takeaway is that removing sensitive attributes did not force the model to find “hidden proxies” for discrimination. Instead, the model shifted its attention to legitimate credit factors. This is a strong practical argument in favour of feature suppression.

3.5. Part 3 Summary

Table 13: Summary of Part 3 findings

Aspect	Finding
Performance	k-Anonymity had the largest utility cost (-0.0478 F1). Feature suppression and DP ($\epsilon = 1.0$) slightly improved F1.
Fairness (sex)	Feature suppression nearly eliminated the sex gap: DP Ratio $0.9084 \rightarrow 0.9950$, TPR gap $0.0600 \rightarrow 0.0150$.
Fairness (age)	Feature suppression improved age fairness (DP Ratio $0.9331 \rightarrow 0.9586$). k-Anonymity worsened it ($0.9331 \rightarrow 0.7876$).
Feature importance	Removed sensitive features had low importance ($\sim 6\%$ combined). Their importance shifted to legitimate financial features (loan purpose, credit history).
Overall	Feature suppression is the best technique for both utility and fairness. It costs almost nothing in accuracy, dramatically improves fairness, and does not create proxy discrimination.

The overall conclusion of Part 3 is that **privacy and fairness reinforce each other**. The simplest privacy technique, feature suppression, turned out to be the most effective fairness intervention as well. It removed the sensitive attributes that the law says should not be used, and the model responded by relying more on legitimate financial facts. This is the ideal outcome for a credit scoring system: the model becomes both more private and more fair, with essentially no cost in predictive accuracy.

4. Part 4: Privacy vs Utility Analysis

In this part, we analyse the trade-offs observed when applying privacy-preserving techniques to our Decision Tree credit scoring model. All results reference the test set of 200 records reported in Table 9 and Table 11.

4.1. What trade-offs did you observe?

The three techniques produced very different trade-offs depending on which privacy threat they targeted.

k-Anonymity ($k=5$) delivered the strongest re-identification protection, reducing the linkage-attack risk from 49.0% to 0% on the training set (Table 8). However, it carried the largest utility cost: suppressing 279 of 800 training records (34.9%) and coarsening `age`, `amount`, and `duration` into broad buckets caused the F1 score to drop by 0.0478, from 0.8199 to 0.7721.

Differential Privacy at $\epsilon = 0.5$ produced a moderate utility cost (F1 drop of 0.0226) because Laplace noise distorted the high-importance numeric columns `status`, `duration`, and `amount`, which together account for over 62% of total Gini importance. At $\epsilon = 1.0$, the noise was mild enough to act as regularisation, reducing overfitting and producing a marginal F1 gain of +0.0040.

Feature suppression removed the five Tier 1 attributes (`sex`, `age`, `age_group`, `personal_status_sex`, `foreign_worker`), which together held only $\sim 6.2\%$ of combined Gini importance in the baseline tree. Removing low-importance columns reduced noise in the feature space, so the F1 score improved by +0.0083, reaching 0.8282. At the same time, the sex Demographic Parity Ratio rose from 0.9084 to 0.9950, and the age group ratio improved from 0.9331 to 0.9586 (Table 12).

4.2. Why does improving privacy reduce model utility?

Privacy techniques reduce utility because they deliberately limit the information available to the model at training time. k-Anonymity achieves this in two ways: it generalises precise numeric values into coarse buckets (for example, grouping `amount` into 1,000 DM intervals and `duration` into 12-month intervals), and it suppresses any record whose quasi-identifier group contains fewer than five members. The result is that the tree trains on 521 records instead of 800, and on rounded rather than exact values for three of its most important features. With less data and lower resolution, the model cannot learn fine-grained decision boundaries, which explains the accuracy drop from 0.7100 to 0.6900.

Differential Privacy at $\epsilon = 0.5$ retains all 800 training records but adds Laplace noise scaled to each column's observed range. For `status_no_checking_account`, which alone carries a Gini importance of 0.3662, even moderate perturbation degrades the signal the tree depends on most, reducing recall from 0.9429 to 0.8286.

Feature suppression is the exception: because the removed attributes had low predictive value, the model did not lose meaningful signal. The freed importance redistributed to legitimate financial features such as `purpose_car (used)` (+0.0266) and `credit_history_critical_account` (+0.0105), which are appropriate credit factors under ECOA and GDPR Article 22.

4.3. Is the trade-off acceptable in your application?

Yes. For a deployed credit scoring system, legal compliance and fairness are non-negotiable constraints, and our results show that meaningful privacy can be achieved with modest utility costs.

The most serious legal risk identified in Part 1 is discrimination: using `sex`, `age`, and `foreign_worker` to influence lending decisions violates the Equal Credit Opportunity Act and GDPR Article 22. Feature suppression eliminates this risk entirely at a net utility gain of +0.0083 F1, making it a strictly dominant choice over the baseline.

k-Anonymity's utility cost (−0.0478 F1, accuracy 0.6900) is larger but still acceptable for a system where a false approval carries real financial risk. Blocking linkage attacks that previously exposed 43.2% of records justifies the performance reduction, particularly given that the technique is applied to stored data rather than live predictions.

Differential Privacy at $\epsilon = 1.0$ adds further protection against membership inference and model-inversion attacks at negligible cost (+0.0040 F1), making it a practical addition to the layered deployment recommended in Part 2.

5. Part 5: Privacy-Aware Deployment Design

This section specifies a concrete deployment plan for the credit scoring system. Each decision is grounded in the privacy risks identified in Part 1 and the technique results from Parts 2 and 3.

5.1. What data should be collected?

The application form should collect only the features that carry legitimate predictive value and are permissible under ECOA and GDPR Article 5(1)(c) (data minimisation). Based on the feature importance analysis in Part 3, the retained features are:

- **Financial status:** `status` (checking account level), `savings`, `credit_history`, `number_credits`, `other_installment_plans`
- **Loan terms:** `amount`, `duration`, `installment_rate`, `purpose`
- **Employment:** `employment_duration` (retained as a financial stability indicator, not as a quasi-identifier in the live model)
- **Obligations:** `other_debtors`, `people_liable`

These columns correspond to Tier 2 (financially sensitive) and Tier 3 (quasi-identifier) attributes from Table 1. They are the features the baseline Decision Tree relies on for over 93% of its total Gini importance.

5.2. What should not be collected?

The five Tier 1 attributes must not appear on the loan application form or in any data pipeline connected to the model:

- `sex` and `personal_status_sex`: protected under ECOA and GDPR

- **age** and **age_group**: protected under ECOA; combined Gini importance only 0.0364 in the baseline, confirming no meaningful predictive loss from removal
- **foreign_worker**: immigration status is a protected characteristic; 96.3% of applicants share the same value, so the column adds no predictive information

If age or identity documents are required by a separate department for fraud verification, that data must be stored in an isolated system with no API connection to the ML pipeline.

5.3. Who can access outputs?

Model outputs are restricted as follows, based on the probability score leakage risk identified in Part 1, Question 2 (Attack 4):

- **Loan officers**: receive only the binary decision (Good / Bad). The continuous probability score is suppressed before the decision reaches any human reviewer, preventing the financial distress inference described in Harm 4.
- **Applicants**: receive the binary outcome and, under GDPR Article 22, a plain-language explanation of the main factors influencing the decision (e.g., checking account status, loan duration).
- **Senior data scientists and compliance auditors**: may access raw probability scores in a secured internal environment solely for model monitoring, fairness auditing, and regulatory reporting. Access is logged and reviewed quarterly.
- **External parties**: no model outputs, scores, or applicant records are shared externally.

5.4. Data storage and retention

- All applicant records are encrypted at rest (AES-256) and in transit (TLS 1.3).
- Before long-term storage, k-anonymity ($k \geq 5$) is applied to the stored dataset using the six quasi-identifier columns identified in Part 2, reducing linkage-attack risk from 49.0% to 0% on the training distribution.
- **Denied applications**: records are deleted 60 days after the legal appeal period expires, consistent with standard banking regulatory requirements.
- **Approved loans**: records are retained for the duration of the loan plus five years after closure, in line with typical banking record-keeping obligations, then permanently and verifiably deleted.
- Query logs and model prediction logs are retained for 12 months for security auditing, then purged.

5.5. Response to privacy breaches

In the event of a confirmed or suspected privacy breach, the following steps are executed in order:

1. **Immediate containment**: the model API is taken offline to halt any ongoing model-inversion or membership inference attacks, as described in Part 1, Question 2.

2. **Scope assessment:** the security team identifies which records were exposed, using the stored k-anonymised audit logs to bound the affected population.
3. **Regulatory notification:** the relevant data protection authority is notified within 72 hours of breach discovery, as required by GDPR Article 33.
4. **Affected individual notification:** if the breach is likely to result in high risk to individuals (e.g., financial fraud or identity theft from exposure of Tier 2 attributes), each affected applicant is notified without undue delay under GDPR Article 34, with a clear description of the data exposed and recommended protective actions.
5. **Remediation and review:** the security vulnerability is closed, the model pipeline is audited, and the incident is documented. The model is not redeployed until the compliance team confirms that all gaps identified in this review have been addressed.